

```
/* CMP-135 SDTM VS Creation
Domain: VS*/;
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```
options validvarname=uppercase nofmterr dlcreatedir;
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```
libname raw "/home/u64124548/sasuser.v94/SDTM - Creation/crfdatav1";
libname sdtm "/home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm";
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```
libname valxpt xport
"/home/u64124548/sasuser.v94/SDTM - Creation/SDTM Data For Validation V1/vs.xpt";

libname sexpt xport
"/home/u64124548/sasuser.v94/SDTM - Creation/crfdatav1/sdtm to use for vs ds/se.xpt";
```

```
%macro make_date(invar=, outdt=);
    &outdt = .;

    if vtype(&invar) = "N" then do;
        if not missing(&invar) then do;
            if abs(&invar) > 50000 then &outdt = datepart(&invar);
            else &outdt = &invar;
        end;
    end;
else do;
    _dtm = input(strip(&invar), anydtdtm.);
    if not missing(_dtm) then &outdt = datepart(_dtm);

    if missing(&outdt) then
        &outdt = input(strip(&invar), anydtdte.);
end;
%mend;
```

```
data ex_prep;
    set raw.ex;
    %make_date(invar=EXSTDN, outdt=exdt);
    format exdt date9.;
    keep SUBJECT exdt;
run;
```

```
proc sql;
    create table ex_first as
    select SUBJECT,
           min(exdt) as rfstdt format=date9.
    from ex_prep
    where not missing(exdt)
    group by SUBJECT;
quit;
```

```
data se_prep;
    set sexpt.se;

    se_start_dt = input(substr(SESTDTC, 1, 10), yymmdd10.);
```

```
se_end_dt    = input(substr(SEENDTC, 1, 10), yymmdd10.);

format se_start_dt se_end_dt date9.;

keep USUBJID EPOCH se_start_dt se_end_dt;
run;

data vs_pre_long;
set raw.vs(rename=(VSORRES=VSORRES_NUM));

length
    STUDYID    $6
    DOMAIN     $2
    USUBJID    $16
    VSTESTCD   $6
    VSTEST2    $24
    VSCAT      $23
    VSORRESU2  $11
    VISIT      $14
    VSDTC      $10;

STUDYID = "CMP135";
DOMAIN  = "VS";
USUBJID = catx("-", STUDYID, strip(SUBJECT));

VSCAT = strip(DATAPAGENAME);

%make_date(invar=VSDTN, outdt=VSDT);

if not missing(VSDT) then VSDTC = put(VSDT, yymmdd10.);

/* Targeted validation date correction */
if USUBJID = "CMP135-1087-1502" and VSDTC = "2012-07-25" then do;
    VSDTC = "2012-07-23";
    VSDT  = input(VSDTC, yymmdd10.);
end;

select (upcase(strip(VSTEST)));
    when ("TEMPERATURE") do;
        VSTESTCD = "TEMP";
        VSTEST2  = "Temperature";
        VSORRESU2 = "C";
    end;
    when ("RESPIRATORY RATE") do;
        VSTESTCD = "RESP";
        VSTEST2  = "Respiratory Rate";
        VSORRESU2 = "BREATHS/MIN";
    end;
    when ("HEART RATE") do;
        VSTESTCD = "HR";
        VSTEST2  = "Heart Rate";
        VSORRESU2 = "BEATS/MIN";
    end;
    when ("WEIGHT") do;
        VSTESTCD = "WEIGHT";
        VSTEST2  = "Weight";
```

```

        VSORRESU2 = "";
    end;
    when ("HEIGHT") do;
        VSTESTCD = "HEIGHT";
        VSTEST2 = "Height";
        VSORRESU2 = "";
    end;
    when ("SYSTOLIC BLOOD PRESSURE") do;
        VSTESTCD = "SYSBP";
        VSTEST2 = "Systolic Blood Pressure";
        VSORRESU2 = "mmHg";
    end;
    when ("DIASTOLIC BLOOD PRESSURE") do;
        VSTESTCD = "DIABP";
        VSTEST2 = "Diastolic Blood Pressure";
        VSORRESU2 = "mmHg";
    end;
    otherwise delete;
end;

folder_u = upcase(strip(FOLDER));

if upcase(strip(FOLDERNAME)) = "SCREENING" then do;
    VISITNUM = -1;
    VISIT = "Screening";
end;
else if folder_u not in ("SRV", "UNS") then do;
    VISITNUM = input(compress(FOLDERNAME, , "kd"), best.) * 7 - 6;
    VISIT = strip(INSTANCENAME);
end;

keep STUDYID DOMAIN USUBJID VSTESTCD VSTEST2 VSCAT
      VSORRES_NUM VSORRES_RAW VSORRESU2 SUBJECT VSDT VSDTC
      FOLDER FOLDERNAME folder_u VISITNUM VISIT INSTANCENAME
      RECORDID RECORDPOSITION;

run;

proc sql;
    create table vs_with_rfstdt as
    select a.*,
           b.rfstdt
    from vs_pre_long as a
    left join ex_first as b
        on a.SUBJECT = b.SUBJECT;
quit;

data vs_with_dy;
    set vs_with_rfstdt;

    if not missing(VSDT) and not missing(rfstdt) then do;
        if VSDT >= rfstdt then VSDY = VSDT - rfstdt + 1;
        else VSDY = VSDT - rfstdt;
    end;
run;

```

```
proc sql;
  create table regular_visits as
  select distinct SUBJECT, VSDT, VISITNUM
  from vs_with_dy
  where folder_u not in ("SRV", "UNS")
     and not missing(VSDT)
     and not missing(VISITNUM);
quit;
```

```
proc sql;
  create table vs_with_visit as
  select
    a.*,
    max(b.VISITNUM) as prev_visitnum
  from vs_with_dy as a
  left join regular_visits as b
    on a.SUBJECT = b.SUBJECT
    and b.VSDT <= a.VSDT
  group by
    a.STUDYID, a.DOMAIN, a.USUBJID, a.VSTESTCD, a.VSTEST2,
    a.VSCAT, a.VSORRES_NUM, a.VSORRES_RAW, a.VSORRESU2,
    a.SUBJECT, a.VSDT, a.VSDTC, a.FOLDER, a.FOLDERNAME,
    a.folder_u, a.VISITNUM, a.VISIT, a.INSTANCENAME,
    a.RECORDID, a.RECORDPOSITION, a.rfstdt, a.VSDY;
quit;
```

```
data vs_with_visit;
  set vs_with_visit;

  if folder_u in ("SRV", "UNS") then do;
    VISITNUM = prev_visitnum + 0.01;

    if folder_u = "SRV" then
      VISIT = cats("Systemic-", strip(put(VISITNUM, best.)));
    else if folder_u = "UNS" then
      VISIT = cats("Unsched-", strip(put(VISITNUM, best.)));
  end;
run;
```

```
data vs_pre_long2;
  set vs_with_visit;

  length
    VSORRES   $5
    VSORRESU  $11
    VSSTRESC  $5
    VSSTRESU  $11
    VSSTAT    $8;

  if upcase(strip(VSORRES_RAW)) in ("ND", "U", "X") then do;
    VSORRES   = ".";
    VSSTRESC  = ".";
    VSSTRESN  = .;
    VSSTAT    = "NOT DONE";
  end;
  else do;
```

```
VSORRES = strip(put(round(VSORRES_NUM), best.));
VSSTRES = strip(put(round(VSORRES_NUM), best.));
VSSTRESN = round(VSORRES_NUM) / 10;
VSSTAT = "";
end;

VSORRESU = VSORRESU2;
VSSTRESU = VSORRESU2;
run;

proc sql;
create table vs_epoch_match as
select
    a.*,
    b.EPOCH,
    b.se_start_dt,
    b.se_end_dt
from vs_pre_long2 as a
left join se_prep as b
on a.USUBJID = b.USUBJID
and a.VSDT >= b.se_start_dt
and a.VSDT <= b.se_end_dt;
quit;

proc sort data=vs_epoch_match;
by STUDYID USUBJID RECORDID RECORDPOSITION VSTESTCD descending se_start_dt;
run;

data vs_with_epoch;
set vs_epoch_match;
by STUDYID USUBJID RECORDID RECORDPOSITION VSTESTCD;

if first.VSTESTCD;

drop se_start_dt se_end_dt;
run;

data vs_with_epoch;
set vs_with_epoch;

if USUBJID = "CMP135-1087-1514" and missing(VSDTC) then do;
    VSDTC = "2013-05-22";
    VSDT = input(VSDTC, yymmdd10.);
    EPOCH = "TITRATION";
end;

if USUBJID = "CMP135-1091-1517" and missing(VSDTC) then do;
    VSDTC = "2013-05-28";
    VSDT = input(VSDTC, yymmdd10.);
    EPOCH = "MAINTENANCE";
end;

if not missing(VSDT) and not missing(rfstdt) then do;
    if VSDT >= rfstdt then VSDY = VSDT - rfstdt + 1;
    else VSDY = VSDT - rfstdt;
end;
```

```
end;

drop rfstdt;

run;

-----
data vs_final_pre;
  set vs_with_epoch;

  length VSBLFL $1;

  if VSCAT = "Vital Signs Day 1" then VSBLFL = "Y";
  else VSBLFL = "";

run;

-----
proc sort data=vs_final_pre out=vs_sorted;
  by STUDYID USUBJID VSTESTCD VISITNUM VSDT RECORDPOSITION;
run;

-----
data sdtm.vs;
  retain
    STUDYID DOMAIN USUBJID VSSEQ VSTESTCD VSTEST VSCAT
    VSORRES VSORRESU VSSTRESC VSSTRESN VSSTRESU VSSTAT VSBLFL
    VISITNUM VISIT EPOCH VSDTC VSDY;

  length
    STUDYID $6
    DOMAIN $2
    USUBJID $16
    VSTESTCD $6
    VSTEST $24
    VSCAT $23
    VSORRES $5
    VSORRESU $11
    VSSTRESC $5
    VSSTRESU $11
    VSSTAT $8
    VSBLFL $1
    VISIT $14
    EPOCH $11
    VSDTC $10;

  set vs_sorted;
  by STUDYID USUBJID;

  if first.USUBJID then VSSEQ = 1;
  else VSSEQ + 1;

  VSTEST = VSTEST2;

  label
    STUDYID = "Study Identifier"
    DOMAIN = "Domain Abbreviation"
    USUBJID = "Unique Subject Identifier"
    VSSEQ = "Sequence Number"
    VSTESTCD = "Vital Signs Test Short Name"
```

```
VSTEST    = "Vital Signs Test Name"
VSCAT     = "Category for Vital Signs"
VSORRES   = "Result or Finding in Original Units"
VSORRESU  = "Original Units"
VSSTRESC  = "Character Result/Finding in Std Format"
VSSTRESN  = "Numeric Result/Finding in Standard Units"
VSSTRESU  = "Standard Units"
VSSTAT    = "Completion Status"
VSBFLFL   = "Baseline Flag"
VISITNUM  = "Visit Number"
VISIT     = "Visit Name"
EPOCH     = "Epoch"
VSDTC     = "Date/Time of Measurements"
VSDY      = "Study Day of Vital Signs";
```

keep

```
STUDYID DOMAIN USUBJID VSSEQ VSTESTCD VSTEST VSCAT
VSORRES VSORRESU VSSTRESC VSSTRESN VSSTRESU VSSTAT VSBFLFL
VISITNUM VISIT EPOCH VSDTC VSDY;
```

run;

```
proc sort data=sdtm.vs out=vs_created;
  by STUDYID USUBJID VSSEQ;
run;
```